Commodities Markets and Models

ESILV

Part I: Market Fundamentals & Spot Price Modeling

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Outline

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1.1 1.1 Introduction to Commodity Markets
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1.1.1 What is a Commodity? (Part A)

Definition: Commodity

A **commodity** is a basic physical good that is:

- Fungible (interchangeable with goods of the same type)
- Used as an input in production or consumption
- Traded on organized exchanges with standardized contracts

Examples: crude oil, wheat, gold, natural gas, copper.

Key Characteristics

- **Homogeneity**: One unit is indistinguishable from another (given grade/quality)
- **Storability**: Most (but not all) commodities can be stored over time
- **Physical delivery**: Contracts may settle via physical delivery or cash
- **Price volatility**: High sensitivity to supply-demand shocks

1.1.1 What is a Commodity? (Part B)

Market Participants & Their Objectives

Participant	Objective	Mathematical Expression
Producers	Secure revenues against price risk	$\max\ \mathbb{E}[P \cdot Q - C(Q)]$
Traders	Exploit arbitrage and spreads	$\max[(P_2 - P_1) \cdot Q - TC]$
End-users	Minimize procurement costs	$\min \int_0^T P(t)\, D(t)\, dt$
Speculators	Maximize risk-adjusted returns	$\max \frac{\mathbb{E}[R]}{\sigma_R}$
Banks	Structure derivative products	$\Pi = \sum_i w_i \cdot \text{Asset}_i$
States	Stabilize fiscal revenues	$\min \text{Var}(\text{Tax Revenue})$

Notation

- $P, P(t)$: commodity price
- Q: quantity
- $C(Q)$: production cost
- TC: transaction costs
- $D(t)$: demand / consumption rate
- R: return
- σ_R: return volatility
- w_i: portfolio weights

1.1.2 Overview of Major Markets (Part A)

Classification of Commodity Markets

Commodities are grouped into broad categories based on their physical nature and use:

- **Energy**: Fuels and power
- **Agricultural**: Food, softs, grains
- **Metals**: Industrial and precious metals
- **Other**: Freight, weather derivatives, emissions

Energy Commodities

- **Crude Oil**: Brent, WTI (West Texas Intermediate)
- **Natural Gas**: Henry Hub (US), NBP (UK), TTF (Europe)
- **Refined Products**: Gasoline, diesel, heating oil
- **Power**: Day-ahead, futures (non-storable)
- **Coal**: Thermal coal, metallurgical coal
- **Emissions**: Carbon credits (EU ETS, voluntary markets)

1.1.2 Overview of Major Markets (Part B)

Category	Examples	Key Exchanges
Energy	Crude oil, natural gas, coal, power, emissions	NYMEX, ICE, NordPool
Agricultural	Wheat, corn, soybeans, coffee, sugar, cotton	CBOT, CME
Metals (Base)	Copper, aluminum, zinc, nickel, lead	LME, SHFE
Metals (Precious)	Gold, silver, platinum, palladium	COMEX, LBMA
Freight	Dry bulk, tanker rates, container	Baltic Exchange
Weather/CAT	Temperature (HDD/CDD), rainfall, hurricane indices	CME

Commoditization Process

As markets mature, products become increasingly standardized:

$$\frac{\partial \text{Premium Margin}}{\partial t} < 0 \quad \text{as} \quad \frac{\partial \text{Standardization}}{\partial t} > 0$$

Crude Oil NYMEX prompt-month price in USD/barrel

Natural Gas NYMEX prompt-month price in USD/MMBtu

UK Gas

London Gas Oil Futures ↓ 611.25 -11.50 (-1.85%)

Buy Sell

Analyze chart

1 Day	1 Week	1 Month	3 Months	6 Months	1 Year	5 years	Max
-1.85%	-1.37%	-6.61%	-8.26%	-12.80%	-13.05%	+45.62%	+98.62%

Wheat CBOT futures price in USD cents/bushel

Corn CBOT futures price in USD cents/bushel

1 Day	1 Week	1 Month	3 Months	6 Months	1 Year	5 years	Max
-0.62%	-2.78%	-2.18%	+4.42%	+0.11%	-2.94%	-9.56%	+45.95%

Soybeans CBOT futures price in USD cents/bushel

Copper LME futures price in USD/Mt

UK Aluminium

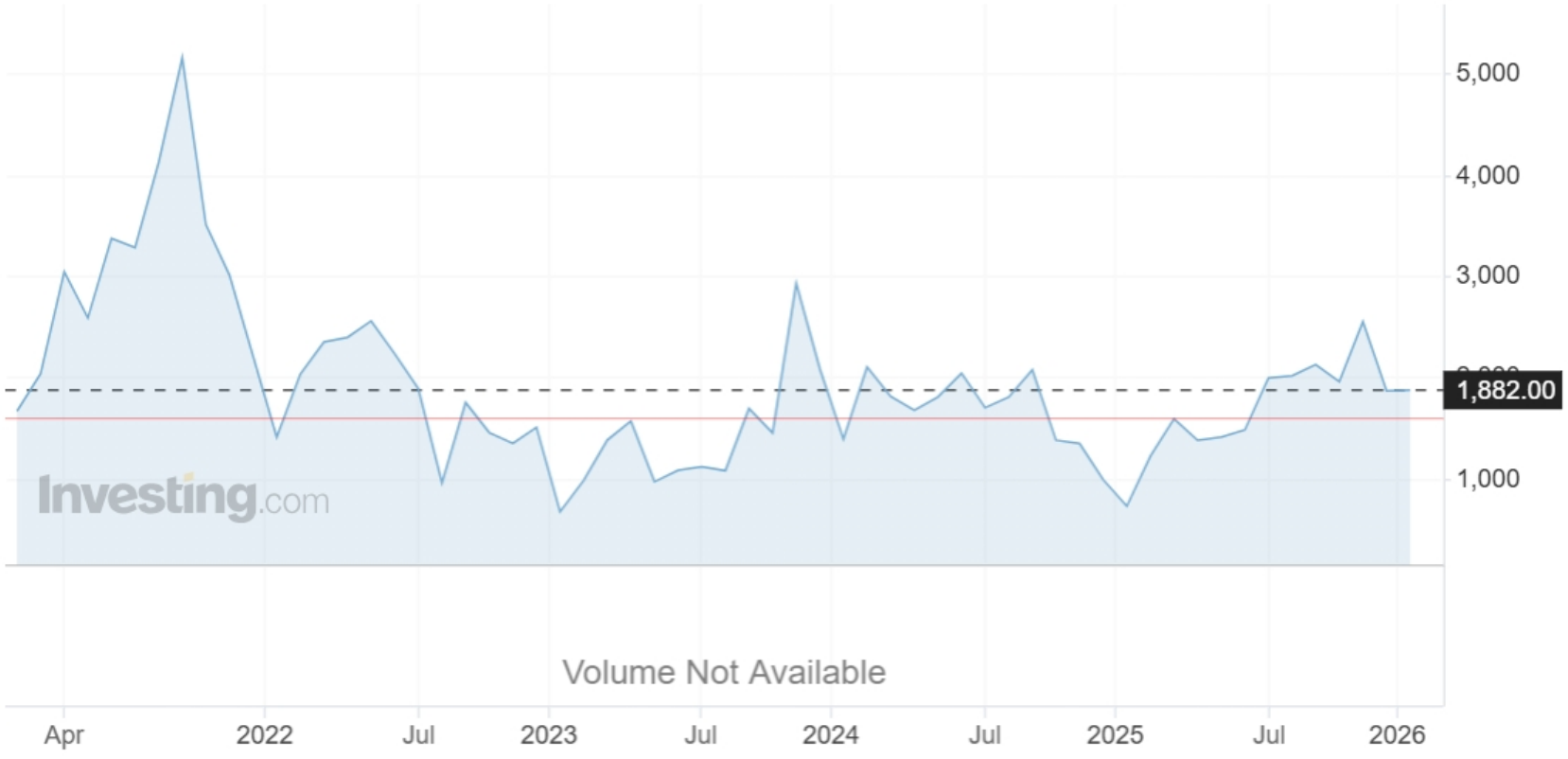
Gold in the London Metal Exchange (USD/Ounce)

Baltic Dry Index

Baltic Dry Index ↑ 1,882.00 +5.00 (+0.27%)

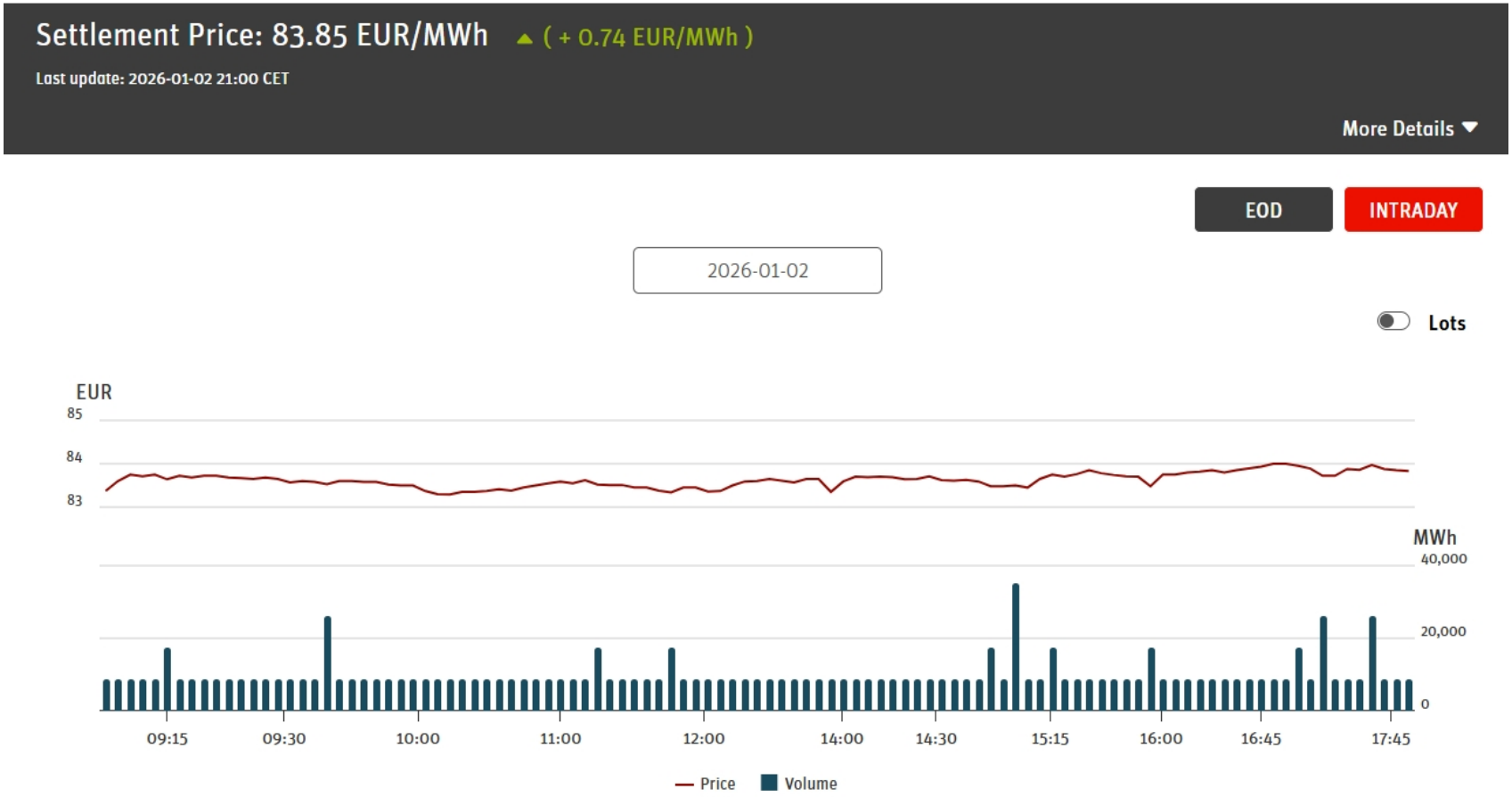
Buy Sell

Analyze chart



1 Day	1 Week	1 Month	3 Months	6 Months	1 Year	5 years	Max
+0.27%	+0.27%	-33.12%	-1.00%	+31.06%	+75.56%	+36.97%	+88.20%

EEX daily spot prices in euro/MWh



[Self-study] Learn how each commodity market works in Helyette Geman's books

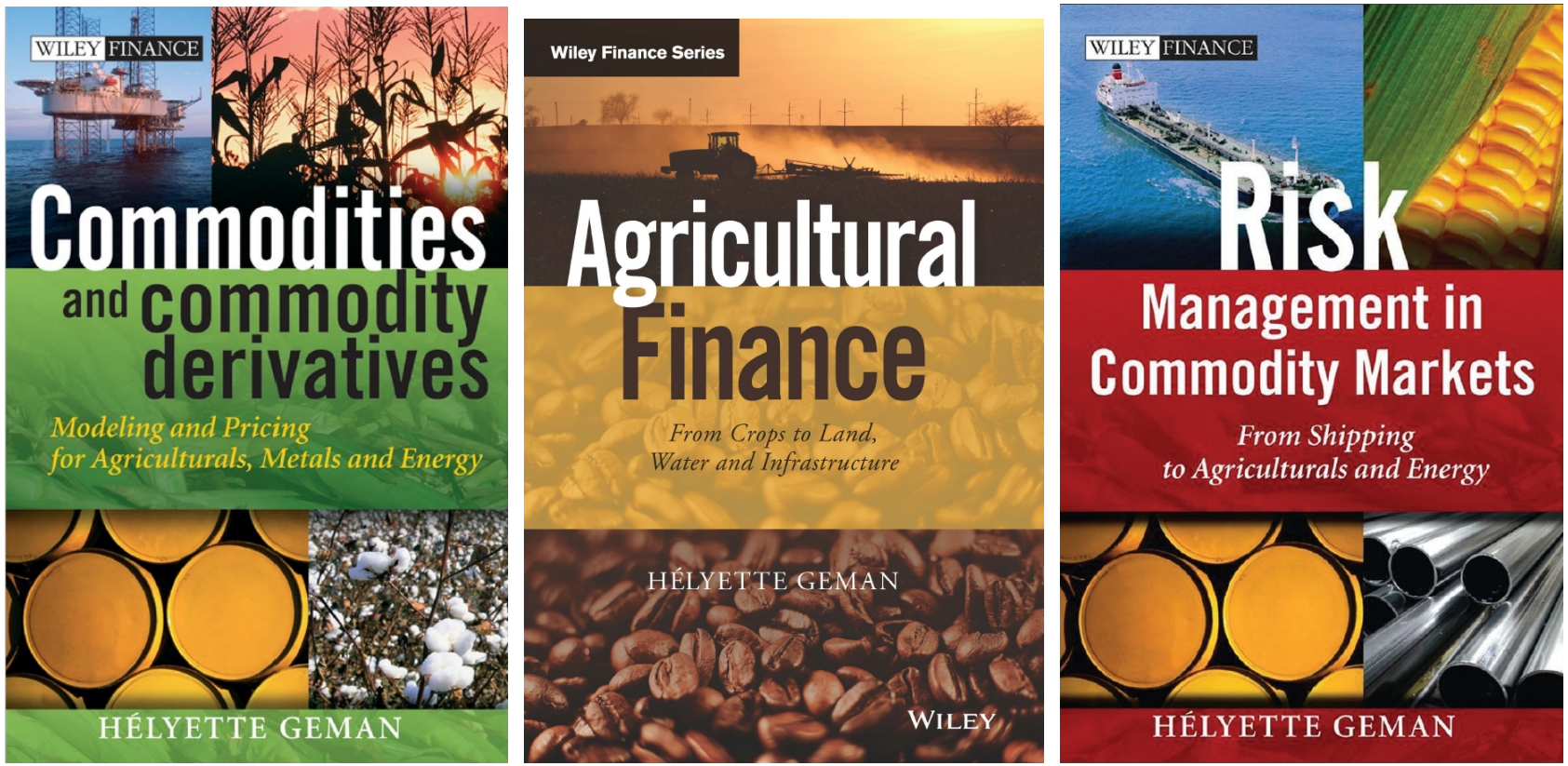


Figure: Find the books on DVL

1.1.3 Key Price Drivers (Part A)

Definition: Supply-Demand Equilibrium

The spot price P_t^* is determined by the intersection of supply and demand curves:

$$P_t^* = \arg\min_P |S(P,t) - D(P,t)|$$

where $S(P,t)$ = supply function, $D(P,t)$ = demand function.

Supply Function

Supply is typically upward-sloping in price:

$$S(P,t) = S_0 + \alpha P + \beta_1 t + \beta_2 W_t + \beta_3 I_{t-1} + \epsilon_t^S$$

where:

- $\alpha > 0$: price elasticity of supply
- W_t: weather/climate factors
- I_{t-1}: inventory levels (previous period)
- ϵ_t^S: supply shocks (e.g., production outages)

1.1.3 Key Price Drivers (Part B)

Demand Function

Demand is typically downward-sloping in price:

$$D(P, t) = D_0 - \gamma P + \delta_1 t + \delta_2 Y_t + \delta_3 T_t + \epsilon_t^D$$

where:

- $\gamma > 0$: price elasticity of demand
- Y_t: GDP or industrial production (business cycle)
- T_t: temperature or seasonal factors
- ϵ_t^D: demand shocks (e.g., extreme weather events)

Equilibrium Condition

At equilibrium: $S(P_t^*, t) = D(P_t^*, t)$

Market clears when quantity supplied equals quantity demanded.

1.1.3 Key Price Drivers (Part C): Storage Theory

Definition: Convenience Yield

The **convenience yield** δ_t represents the benefit of physically holding a commodity (e.g., avoiding stockouts, operational flexibility). It is implicit in futures prices.

Storage Theory (Working, 1949)

The relationship between spot S_t and futures $F(t, T)$ prices is:

$$F(t, T) = S_t \exp\left[(r + s - \delta_t)(T - t)\right]$$

where:

- r = risk-free interest rate
- s = storage cost (% per annum)
- δ_t = convenience yield (% per annum)

1.1.3 Key Price Drivers (Part D): Convenience Yield

Modeling Convenience Yield

Convenience yield is a decreasing function of inventory levels:

$$\delta_t = \delta(I_t) = \alpha_0 + \alpha_1 I_t + \alpha_2 I_t^2$$

where $\alpha_1 < 0$ (higher inventories $\Rightarrow$ lower convenience yield).

Implications

- **High inventories** (I_t large): δ_t small $\Rightarrow$ futures trade at premium to spot (**contango**)
- **Low inventories** (I_t small): δ_t large $\Rightarrow$ spot trades at premium to futures (**backwardation**)

This is the **Theory of Storage** linking physical inventories to forward curve shape.

1.1.3 Key Price Drivers (Part E): Super-Cycles

Definition: Commodity Super-Cycle

A **super-cycle** is a prolonged period (10-30 years) of elevated commodity prices driven by structural demand shifts (e.g., industrialization of China, energy transitions).

Structural Decomposition Model

Commodity prices can be decomposed into trend, cycles, and noise:

$$P_t = \underbrace{\mu_t}_{\text{Long-term trend}} + \underbrace{\sum_{k=1}^{K} \left[\phi_k \cos\left(\frac{2\pi kt}{L}\right) + \psi_k \sin\left(\frac{2\pi kt}{L}\right) \right]}_{\text{Cyclical components}} + \underbrace{\epsilon_t}_{\text{Short-term noise}}$$

where L is the cycle length (e.g., 20 years for super-cycle).

1.1.3 Key Price Drivers (Part F): Trend Dynamics

Mean-Reverting Trend

The long-term trend μ_t itself evolves stochastically:

$$d\mu_t = \kappa(\theta - \mu_t)dt + \sigma_\mu dW_t^\mu$$

where:

- $\kappa > 0$: speed of mean-reversion
- θ: long-run equilibrium level
- σ_μ: volatility of trend
- W_t^μ: Brownian motion

Hodrick-Prescott Filter

To extract the trend τ_t from observed prices P_t:

$$\min_{\{\tau_t\}} \sum_{t=1}^{T} (P_t - \tau_t)^2 + \lambda \sum_{t=2}^{T-1} \left[(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1})\right]^2$$

Large λ (e.g., 100,000) isolates low-frequency components (super-cycles).

1.2.1 Spot, Forwards, Futures, and Swaps (Part A)

Definition: Spot Price

The **spot price** S_t is the price for immediate (or near-immediate) delivery of a commodity at a specified:

- Location (L): e.g., Cushing, OK for WTI crude
- Grade (G): e.g., quality specifications
- Time (T): typically next business day or within 2-3 days

Notation: $S_t = S_t(L, G, T)$

Remark: Physical vs. Financial Settlement

- **Physical delivery**: Actual commodity transferred
- **Cash settlement**: Financial payment based on index price

Most exchange-traded contracts allow cash settlement to avoid logistics.

1.2.1 Spot, Forwards, Futures, and Swaps (Part B)

Definition: Forward Contract

A **forward** is an OTC agreement to buy/sell a commodity at price $F(t, T)$ at future date T, agreed upon at time t. No cash flows until maturity.

Forward Pricing: Cost-of-Carry Model

Under no-arbitrage, the forward price is:

$$F(t, T) = S_t \exp\left[(r + s - y)(T - t)\right]$$

where:

- r = risk-free rate (financing cost)
- s = storage cost (% per annum)
- y = convenience yield (% per annum)

Intuition: Forward price compensates for financing and storage costs, adjusted for benefits of holding the physical commodity.

1.2.1 Spot, Forwards, Futures, and Swaps (Part C)

Definition: Futures Contract

A **futures** contract is an exchange-traded, standardized agreement to buy/sell at a future date. Key features:

- Daily mark-to-market (margin calls)
- Centralized clearing (counterparty risk mitigation)
- Standardized specifications

Futures vs. Forward Prices

Due to mark-to-market and interest rate correlation:

$$F^{\text{fut}}(t, T) = F^{\text{for}}(t, T) + \text{Convexity Adjustment}$$

For uncorrelated interest rates and commodity prices:

$$\text{Convexity Adjustment} \approx \frac{1}{2}\sigma^2(T-t)^2 \cdot \text{Cov}(r, S)$$

In practice, this adjustment is often negligible for short maturities.

1.2.1 Spot, Forwards, Futures, and Swaps (Part D)

Definition: Commodity Swap

A **commodity swap** exchanges floating spot prices for a fixed price over multiple periods. Used for long-term hedging.

Swap Pricing Formula

The fair swap rate is the weighted average of forward prices:

$$SW(t; T_1, \ldots, T_N) = \frac{\sum_{i=1}^{N} D(t, T_i) F(t, T_i)}{\sum_{i=1}^{N} D(t, T_i)}$$

where:

- $D(t, T) = \exp\left(-\int_t^T r(s)ds\right)$: discount factor
- $F(t, T_i)$: forward price for settlement i

Swap rate SW is the fixed price that makes initial swap value zero.

Basis Swap Payoff

A **basis swap** exchanges two commodity prices (different locations or grades). Payoff from the perspective of receiving S^A and paying $S^B + K$:

$$CF_i = Q \times \left[S^A_{T_i} - S^B_{T_i} - K\right]$$

where:

- S^A: Price at location/grade A (received)
- S^B: Price at location/grade B (paid)
- K: Fixed basis spread
- Q: Notional quantity

Example

WTI–Brent basis swap: Receive Brent, pay WTI plus fixed spread. Used to hedge exposure to price differentials between markets.

[Self-study] Options: Read if you need to catch up

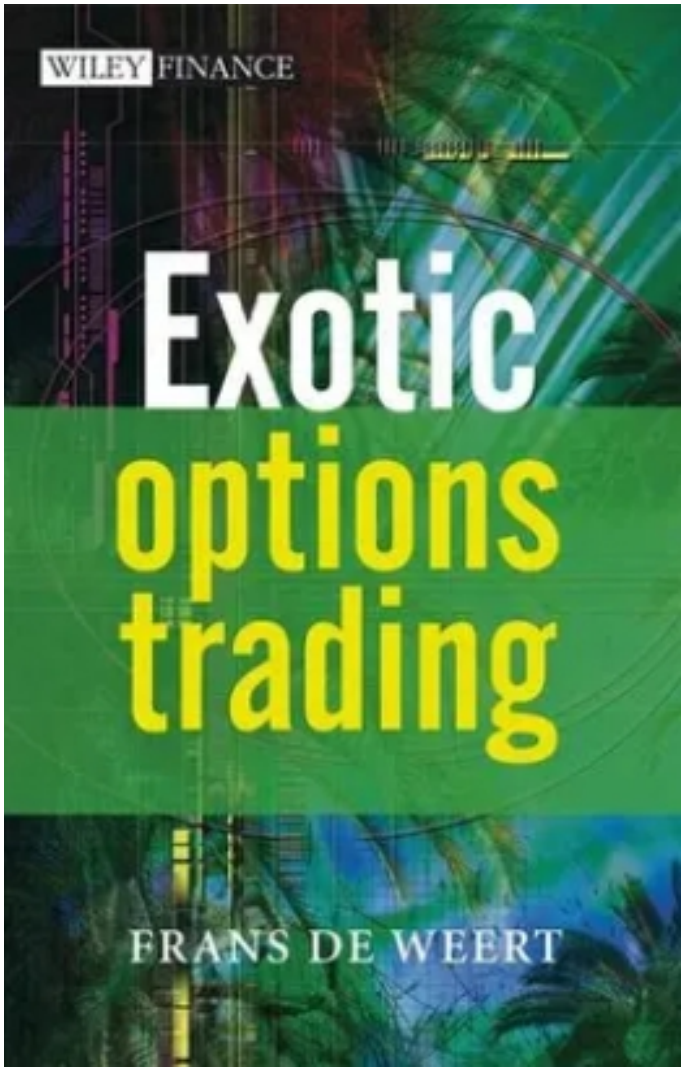


Figure: Find the book on DVL

1.2.2 Options on Spot and Futures (Part A)

Definition: European Call Option

A **call option** gives the holder the *right* (not obligation) to buy the underlying at strike K at maturity T.

Payoff at maturity: $\max(S_T - K, 0)$ for spot option, or $\max(F_T - K, 0)$ for futures option.

Black-76 Model for Futures Options

For a European call on a futures contract:

$$C = e^{-r(T-t)}\left[F(t,T)N(d_1) - KN(d_2)\right]$$

where:

$$d_1 = \frac{\ln(F(t,T)/K) + \frac{1}{2}\sigma^2(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}$$

and $N(\cdot)$ is the cumulative standard normal distribution.

1.2.2 Options on Spot and Futures (Part B)

European Put Option

For a put option (right to sell at K):

$$P = e^{-r(T-t)}\left[KN(-d_2) - F(t,T)N(-d_1)\right]$$

Put-Call Parity for futures options:

$$C - P = e^{-r(T-t)}\left[F(t,T) - K\right]$$

Key Parameters

- $F(t,T)$: Futures price (forward price for cash-settled)
- K: Strike price
- σ: Implied volatility (constant in Black-76)
- r: Risk-free rate
- $T - t$: Time to maturity

1.2.2 Options on Spot and Futures (Part C): Asian Options

Asian Option (Arithmetic Average)

Payoff depends on the average price:

$$\text{Call: } \max\left(\frac{1}{n}\sum_{i=1}^{n} S_{t_i} - K, 0\right)$$

Used in commodities to hedge average production/consumption.

Turnbull-Wakeman Approximation

$$C \approx e^{-rT}\left[M_1 N(d_1) - K N(d_2)\right], \quad d_1 = \frac{\ln(M_1/K) + \frac{1}{2}\sigma_A^2 T}{\sigma_A\sqrt{T}}, \; d_2 = d_1 - \sigma_A\sqrt{T}$$

$$\sigma_A \approx \frac{\sigma}{\sqrt{3}} \text{ (continuous averaging)}$$

Asian options are cheaper than vanilla due to volatility reduction from averaging.

1.2.2 Options on Spot and Futures (Part D): Spread Options

Definition: Spread Option

A **spread option** has payoff based on the difference between two prices:

$$\text{Payoff} = \max(S_1(T) - S_2(T) - K, 0)$$

Common in commodity markets: crack spreads (refining: oil $\rightarrow$ gasoline), spark spreads (power generation: gas $\rightarrow$ electricity), calendar spreads.

Kirk's Approximation (1995)

For spread $S_1 - S_2$ with strike K:

$$C \approx e^{-rT}\left[F_1 N(d_1) - (F_2 + K)N(d_2)\right]$$

$$d_1 = \frac{\ln\left(\frac{F_1}{F_2+K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

1.2.2 Options on Spot and Futures (Part E): Spread Options (cont.)

Effective Volatility for Spread

$$\sigma = \sqrt{\sigma_1^2 + \left(\frac{F_2}{F_2 + K}\right)^2 \sigma_2^2 - 2\rho\sigma_1\sigma_2 \left(\frac{F_2}{F_2 + K}\right)}$$

- σ_1, σ_2: volatilities of S_1, S_2
- ρ: correlation between S_1 and S_2
- F_1, F_2: forward prices
- **Key insight**: higher $\rho \to$ lower spread volatility $\to$ cheaper option

Correlation ρ	Spread volatility	Option cost
High ($\approx$1)	Low	Cheap
Medium ($\approx$0.5)	Medium	Medium
Low ($\approx$0)	High	Expensive

1.2.2 Options on Spot and Futures (Part F1): Margrabe's Formula

Exchange Option (K=0)

Right to exchange one commodity for another at maturity:

$$\text{Payoff} = \max(S_1(T) - S_2(T), 0)$$

Example: Brent vs WTI, corn vs wheat.

Margrabe's Formula

$$C = S_1 e^{-q_1 T} N(d_1) - S_2 e^{-q_2 T} N(d_2)$$

$$d_1 = \frac{\ln(S_1/S_2) + (q_2 - q_1 + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

$$\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$$

1.2.2 Options on Spot and Futures (Part F2): Margrabe's Formula (Intuition)

Intuition

- Pays if $S_1 > S_2$; acts as "insurance" on relative value
- High correlation (ρ close to 1) $\rightarrow$ difference stable $\rightarrow$ cheap option
- Low correlation (ρ close to 0) $\rightarrow$ difference jumps $\rightarrow$ expensive option

Correlation ρ	Spread volatility	Option cost
High ($\approx$1)	Low	Cheap
Medium ($\approx$0.5)	Medium	Medium
Low ($\approx$0)	High	Expensive

1.2.2 Options on Spot and Futures (Part G): Barrier Options

Definition: Barrier Option

A **barrier option** is activated or deactivated when the underlying crosses a barrier level H.

- **Knock-in**: Option activates when S crosses H
- **Knock-out**: Option deactivates when S crosses H

Common types: down-and-out call, up-and-out put, etc.

Down-and-Out Call

For barrier $H < S_0$ (option knocked out if S falls below H):

$$C_{\text{do}} = C_{\text{BS}} - \left(\frac{H}{S_0}\right)^{2\lambda-2} C_{\text{BS}}\left(\frac{H^2}{S_0}, K\right)$$

where $\lambda = \frac{r-q+\frac{1}{2}\sigma^2}{\sigma^2}$ and C_{BS} is Black-Scholes price.

1.2.2 Options on Spot and Futures (Part H): Barrier Interpretation

Interpretation of Barrier Formula

The down-and-out call price is:

$$C_{\text{do}} = \underbrace{C_{\text{BS}}}_{\text{Vanilla call}} - \underbrace{\left(\frac{H}{S_0}\right)^{2\lambda-2} C_{\text{BS}}\left(\frac{H^2}{S_0}, K\right)}_{\text{Knock-out probability adjustment}}$$

- First term: Value if barrier never hit
- Second term: Rebate for paths that hit barrier

Remark

Barrier options are cheaper than vanilla options and widely used for cost reduction in structured products.

1.2.2 Options on Spot and Futures (Part I): Barrier Option Intuition

Barrier Option Intuition

- Barrier options = "vanilla options with a condition": payoff exists only if underlying respects the barrier.
- Down-and-out call: knocked out if $S < H \rightarrow$ cheaper than vanilla call.
- Knock-in option: only activates if barrier is hit $\rightarrow$ can be combined with vanilla for replication.
- Barrier level H controls cost:

Barrier H	Likelihood of knock-out	Approx. option cost
$H \approx S_0$	High	Low (very cheap)
$H < S_0$ but far	Medium	Medium
$H \ll S_0$	Low	$\approx$ Vanilla price

- Intuition: barrier = conditional "safety net" or trigger for payout

1.2.3 Hedging with Futures (Part A)

Definition: Hedging

Hedging is the practice of taking an offsetting position in derivatives to reduce price risk exposure.

Objective: Minimize variance of hedged portfolio, not necessarily maximize expected return.

Minimum Variance Hedge Ratio

The optimal hedge ratio h^* minimizes the variance of the hedged position:

$$h^* = \arg\min_h \mathrm{Var}(\Delta S - h\Delta F)$$

Taking the derivative and setting to zero:

$$h^* = \frac{\mathrm{Cov}(\Delta S, \Delta F)}{\mathrm{Var}(\Delta F)} = \rho_{S,F}\frac{\sigma_S}{\sigma_F}$$

1.2.3 Hedging with Futures (Part B)

Estimation via OLS Regression

The hedge ratio is estimated by regressing spot price changes on futures price changes:

$$\Delta S_t = \alpha + h\Delta F_t + \epsilon_t$$

The OLS estimator $\hat{h}$ equals the optimal hedge ratio h^*.

Interpretation of Components

- $\rho_{S,F}$: Correlation between spot and futures (typically close to 1)
- $\frac{\sigma_S}{\sigma_F}$: Ratio of volatilities
- If $\sigma_S = \sigma_F$ and $\rho = 1$: $h^* = 1$ (perfect hedge)
- If $\rho < 1$: Under-hedge to account for imperfect correlation

1.2.3 Hedging with Futures (Part C): Hedging Effectiveness

Definition: Hedging Effectiveness

Hedging effectiveness measures the proportion of variance eliminated by the hedge:

$$R^2_{\text{hedge}} = \rho^2_{S,F} = \frac{[\text{Cov}(\Delta S, \Delta F)]^2}{\text{Var}(\Delta S)\text{Var}(\Delta F)}$$

This is the R^2 from the regression $\Delta S_t = \alpha + h\Delta F_t + \epsilon_t$.

Variance Reduction

The ratio of hedged to unhedged variance:

$$\frac{\text{Var}(\Delta S - h^* \Delta F)}{\text{Var}(\Delta S)} = 1 - R^2_{\text{hedge}}$$

Example: If $R^2 = 0.90$, the hedge eliminates 90% of price variance.

1.2.3 Hedging with Futures (Part D): Basis Risk

Definition: Basis Risk

Basis risk is the risk that spot and futures prices do not move in perfect correlation. The basis is:

$$B_t = S_t - F(t, T)$$

At maturity T: $B_T = 0$ (convergence), but B_t varies before maturity.

Sources of Basis Risk

1. **Quality basis**: Physical commodity differs from futures contract specification
2. **Location basis**: Delivery location differs from where commodity is produced/consumed
3. **Calendar basis**: Hedge maturity mismatches with actual exposure date

1.2.3 Hedging with Futures (Part E): Basis Decomposition

Three-Component Basis Model

Total basis can be decomposed:

$$B_t = S'_t - F(t, T)$$

$$= \underbrace{[S'_t - S^*_t]}_{\text{Quality basis}} + \underbrace{[S^*_t - F(t, T^*)]}_{\text{Location basis}} + \underbrace{[F(t, T^*) - F(t, T)]}_{\text{Calendar basis}}$$

where:

- S'_t: Actual spot price (your location/grade)
- S^*_t: Benchmark spot price (futures delivery point/grade)
- T^*: Nearest futures contract maturity
- T: Your exposure date

1.2.3 Hedging with Futures (Part F): Basis Dynamics

Mean-Reverting Basis Model

The basis typically follows a mean-reverting process:

$$dB_t = \kappa_B(\theta_B - B_t)dt + \sigma_B dW_t^B$$

where:

- $\kappa_B > 0$: Speed of mean-reversion (basis tends toward zero as maturity approaches)
- θ_B: Long-run average basis
- σ_B: Basis volatility

Variance of Hedged Position

The residual variance after hedging is:

$$\mathrm{Var}(S'_T - F(T,T)) = \mathrm{Var}(B_T) = \frac{\sigma_B^2}{2\kappa_B}\left(1 - e^{-2\kappa_B(T-t)}\right)$$

As $T \to t$: $\mathrm{Var}(B_T) \to 0$ (convergence reduces basis risk).

1.3.1 Empirical Analysis of Price Trajectories (Part A)

Stylized Facts of Commodity Prices

Commodity prices exhibit distinct statistical properties:

- **Mean-reversion**: Prices tend to revert to long-run equilibrium levels
- **Spikes**: Sudden, asymmetric jumps (especially in non-storable commodities)
- **Seasonality**: Predictable periodic patterns
- **Volatility clustering**: High volatility persists (GARCH effects)
- **Fat tails**: Extreme events more frequent than normal distribution predicts

Contrast with Financial Assets

Unlike stocks (often modeled as geometric Brownian motion), commodity prices:

- Do NOT follow random walks (mean-reversion dominates)
- Have stronger seasonality
- Exhibit more pronounced jumps/spikes

1.3.1 Empirical Analysis of Price Trajectories (Part B)

Price Spikes: Characteristics

- **Asymmetry**: Upward spikes larger than downward moves
- **Fast reversion**: Prices quickly return to normal after spike
- **Physical constraints**: Supply inelasticity (cannot produce instantly)

Example: Electricity prices can spike 100x during peak demand, then normalize within hours.

Heteroscedasticity

Volatility σ_t is time-varying:

$$\mathrm{Var}(r_t|\mathcal{F}_{t-1}) = \sigma_t^2 \neq \text{constant}$$

Periods of high volatility cluster together (ARCH/GARCH effects).

1.3.1 Empirical Analysis of Price Trajectories (Part C)

Fat Tails: Kurtosis Analysis

For normally distributed returns: Kurtosis $= 3$

Commodity returns typically show:

$$\text{Kurtosis} = \frac{E[(r-\mu)^4]}{\sigma^4} > 3 \quad \text{(leptokurtic)}$$

This indicates higher probability of extreme events than Gaussian assumption.

Jarque-Bera Test for Normality

Test statistic:

$$JB = \frac{n}{6}\left[S^2 + \frac{(K-3)^2}{4}\right] \sim \chi^2(2)$$

where S = skewness, K = kurtosis, n = sample size.

Result: Commodity returns typically reject normality at high significance levels.

1.3.1 Empirical Analysis of Price Trajectories (Part D)

Forward Curve Slope

$$\text{Slope}_t = \ln(S_t) - \ln\left(F_t^{12M}\right)$$

where S_t is the spot price and F_t^{12M} the 12-month forward price.
Log-difference $\approx$ relative price gap.

Interpretation

- $\text{Slope}_t > 0 \rightarrow$ **Backwardation** (spot $>$ forward $\rightarrow$ tight supply, low inventories)
- $\text{Slope}_t < 0 \rightarrow$ **Contango** (spot $<$ forward $\rightarrow$ ample supply, high inventories)

Mean-Reverting Market Tightness

$$d(\text{Slope}_t) = \kappa(\theta - \text{Slope}_t)\, dt + \sigma\, dW_t$$

Empirically: $\kappa \approx$ 1–3 per year $\rightarrow$ half-life $\approx$ 3–8 months.

1.3.2 Seasonality, Spikes, Mean-Reversion (Part A): Seasonality

Definition: Seasonality

Seasonality = predictable, periodic price fluctuations driven by:

- Weather patterns (heating/cooling demand)
- Agricultural cycles (planting/harvesting)
- Holiday consumption

Why Deseasonalize?

- Improves stationarity tests (seasonal patterns can mask mean-reversion)
- Allows cleaner modeling of stochastic component
- Better volatility estimation

Seasonal Pattern Examples

- **Natural gas**: Peak in winter (heating demand), trough in spring/fall
- **Electricity**: Peak in summer (air conditioning) and winter (heating)
- **Agricultural**: Post-harvest price declines, pre-harvest increases

Deseasonalization Procedure

Step 1: Average log-price by season (day/week/month):

$$\bar{L}_d = \frac{1}{N_d} \sum_{t \in \text{season } d} \log(S_t)$$

Step 2: Compute seasonal factors:

$$\pi_d = \exp\left(\bar{L}_d - \bar{L}\right), \quad \bar{L} = \frac{1}{D} \sum_{d=1}^{D} \bar{L}_d$$

Step 3: Deseasonalize observed prices:

$$\tilde{S}_t = \frac{S_t}{\pi_{d(t)}}$$

Notation:

- S_t: observed spot price at time t
- $\tilde{S}_t$: deseasonalized price
- $d(t)$: season of observation t
- N_d: number of observations in season d
- D: total number of seasons

1.3.2 Seasonality, Spikes, Mean-Reversion (Part C)

Fourier Decomposition

Seasonal component modeled as sum of sinusoids:

$$\log(S_t) = \alpha_0 + \sum_{k=1}^{K} [\alpha_k \cos(2\pi f_k t) + \beta_k \sin(2\pi f_k t)] + \epsilon_t$$

Symbol	**Meaning**
S_t	Spot price at time t
α_0	Mean log-price
α_k, β_k	Fourier coefficients for seasonal component k
$f_k = k/L$	Frequency of season k
L	Period length (e.g., 365 for yearly, 7 for weekly)
ϵ_t	Residual (stochastic) component

Example: Daily natural gas prices $L = 365$ (yearly) or $L = 7$ (weekly).

1.3.2 Seasonality, Spikes, Mean-Reversion (Part D1): Hotelling's Rule + Solution

Hotelling's Rule (1931)

For **exhaustible resources** (oil, minerals) with extraction cost c:

$$\frac{d}{dt}[P_t - c] = r\,[P_t - c] \quad (P_t = \text{spot price at time } t,\ c = \text{marginal extraction cost},\ r = \text{r}$$

Solution (exponential growth of scarcity rent):
This is a first-order linear ODE. Separate variables:

$$\frac{d(P_t - c)}{P_t - c} = r\,dt \quad \text{(instantaneous growth rate of scarcity rent)}$$

Integrate both sides:

$$\ln(P_t - c) = rt + \text{constant} \quad \Rightarrow \quad P_t - c = (P_0 - c)e^{rt} \quad (P_0 = \text{initial spot price})$$

$$\boxed{P_t = c + (P_0 - c)e^{rt}} \quad \text{(spot price grows with scarcity rent)}$$

Note: Scarcity rent $(P_t - c)$ represents the "extra value" of holding the resource for the future.

1.3.2 Seasonality, Spikes, Mean-Reversion (Part D2): Implications & Reality

Commentary

- **Intuition**: Scarcity rent grows exponentially at the risk-free rate if markets are perfectly efficient.
- **Interpretation**: $P_t - c$ represents the "extra value" of holding the resource for the future.
- **Reality check**: Smooth exponential growth rarely occurs due to market frictions.

Empirical Reality

- **Technological progress**: new extraction methods reduce scarcity effect.
- **Discovery of new reserves**: expands supply, lowering scarcity rent.
- **Substitution effects**: alternative resources reduce price pressure.
- **Mean-reversion dominates**: observable prices often revert to a long-term level instead of following pure Hotelling growth.
- **Takeaway**: Hotelling provides a neat theoretical benchmark, but real commodity prices are noisy and mean-reverting.

1.3.2 Seasonality, Spikes, Mean-Reversion (Part E): Storage Mean-Reversion

Storage Arbitrage and Spot Price Dynamics

No-arbitrage condition for storage:

$$F(t, T) - S_t \leq (r + s)(T - t)S_t$$

where:
S_t = spot price, $F(t, T)$ = forward price, r = risk-free rate, s = storage cost
With convenience yield y:

$$S_t e^{(r+s)(T-t)} \geq F(t, T) \geq S_t e^{(r+s-y)(T-t)}$$

Step-by-step intuition for mean-reversion:

- **If spot is high**: market releases inventory $\Rightarrow$ supply rises $\Rightarrow$ prices fall.
- **If spot is low**: market builds inventory $\Rightarrow$ demand rises $\Rightarrow$ prices increase.
- **Effect**: Spot prices are "pulled" toward levels consistent with forward bounds, creating mean-reversion.
- **Takeaway**: Storage + forward constraints naturally generate mean-reverting behavior in commodities.

1.3.2 Seasonality, Spikes, Mean-Reversion (Part F)

Mean-Reversion Speed Estimation

Assume AR(1) model for deviation from long-run level ($x_t = S_t - \bar{S}$):

$$x_{t+1} = \phi x_t + \epsilon_t \quad (\phi = \text{autocorrelation}, \ \epsilon_t = \text{noise})$$

Mean-reversion time constant:

$$\tau = -\frac{1}{\ln \phi}$$

For $\phi \approx 1$ (weak mean-reversion):

$$\tau \approx \frac{1}{1-\phi}$$

Half-life (time to close half the gap):

$$t_{1/2} = \tau \ln 2 = -\frac{\ln 2}{\ln \phi}$$

Example

If $\phi = 0.95$, then $\tau \approx 20$ periods and $t_{1/2} \approx 14$ periods.

1.3.2 Seasonality, Spikes, Mean-Reversion (Part G): Stationarity Tests

Augmented Dickey-Fuller (ADF) Test

Tests null hypothesis of unit root (non-stationarity) vs. stationarity:

$$\Delta x_t = \alpha x_{t-1} + \sum_{i=1}^{p} \beta_i \Delta x_{t-i} + \gamma + \delta t + \epsilon_t$$

Notation:

- x_t: Series under test
- p: number of lags for autocorrelation control
- γ: intercept, δt: deterministic trend
- ϵ_t: residual (white noise)

- $H_0 : \alpha = 0$ (unit root, non-stationary)
- $H_1 : \alpha < 0$ (stationary, mean-reverting)

Test statistic compared to Fuller (1976) critical values (non-standard distribution).

1.3.2 Seasonality, Spikes, Mean-Reversion (Part H): Phillips-Perron Test

PP Test

Non-parametric alternative to ADF:

$$\Delta x_t = \alpha x_{t-1} + \gamma + \delta t + u_t$$

Notation:

- x_t: Series under test
- u_t: residual
- γ, δt: intercept and trend

Modified t-statistic accounts for autocorrelation and heteroscedasticity:

$$Z_t = t_\alpha \left(\frac{\hat{\sigma}^2}{\hat{\lambda}^2} \right)^{1/2} - \frac{(\hat{\lambda}^2 - \hat{\sigma}^2) T}{2\hat{\lambda}\hat{s}}$$

where $\hat{\sigma}^2$ and $\hat{\lambda}^2$ are short-run and long-run variance estimates.

Advantage: Robust to heteroscedasticity and serial correlation without specifying lag structure.

1.3.2 Seasonality, Spikes, Mean-Reversion (Part I): Empirical Results

Series	ADF Stat	p-value	PP Stat	Conclusion
PXT log price	-3.94	0.012	-5.62	**Stationary**
EEX log price	-3.61	0.032	-6.43	**Stationary**
NG Henry Hub spot	-3.05	0.13	-3.06	Non-stationary
WTI crude spot	-2.72	0.27	-2.82	Non-stationary
NG 12M futures	-2.00	0.58	-2.23	Non-stationary
WTI 12M futures	-1.18	0.91	-1.37	Non-stationary
NG slope (spot-12M)	-4.15	< 0.01	-4.87	**Stationary**
WTI slope (spot-12M)	-4.11	< 0.01	-4.09	**Stationary**

Table: Stationarity tests for commodity price series (critical value ≈ -3.4 at 5%)

1.3.2 Seasonality, Spikes, Mean-Reversion (Part J): Interpretation

Empirical Interpretation

1. **Electricity prices (PXT, EEX)**: Stationary
   - Non-storable $\Rightarrow$ no inventory effects
   - Strong mean-reversion to production costs
2. **Storable commodities (NG, WTI) spot/futures**: Non-stationary
   - Long-term trends dominate (demand growth, depletion)
   - Hotelling-like behavior over long horizons
3. **Slope (spot vs. 12M futures)**: Strongly stationary
   - Basis mean-reverts as maturity approaches
   - Storage arbitrage enforces bounds

Practical Implication

Modeling guideline: Spot prices include drift; slopes can be modeled as pure mean-reversion (no drift).

1.4.1 Limitations of Simple Mean-Reversion (Part A)

AR(1) Model (Discrete-Time)

$$x_{t+1} = \phi x_t + \epsilon_{t+1}, \quad \epsilon_t \sim N(0, \sigma^2)$$

Notation: $x_t = \log(S_t) - \mu$ (deviation from long-run mean)

Equivalent form highlighting mean-reversion:

$$x_{t+1} - x_t = (\phi - 1)x_t + \epsilon_{t+1} \quad \Longrightarrow \quad \Delta x_t = -\kappa_d x_t + \epsilon_{t+1}, \quad \kappa_d = 1 - \phi$$

Ornstein-Uhlenbeck (Continuous-Time)

$$dx_t = -\kappa x_t dt + \sigma dW_t$$

Interpretation:

- $-\kappa x_t$ = restoring force toward mean (negative sign essential)
- σdW_t = random shocks (Brownian motion)
- OU process = continuous-time analog of AR(1)

1.4.1 Limitations of Simple Mean-Reversion (Part A, cont.)

Pedagogical clarification:

- AR(1) is discrete-time: no explicit negative sign; mean-reversion encoded by $0 < \phi < 1$
- OU is continuous-time: negative sign ensures pull back to mean
- Both capture the same economic intuition: deviations decay over time

Assumptions (Both AR(1) & OU)

- Gaussian innovations
- Constant volatility (homoscedasticity)
- Linear mean-reversion
- No jumps

1.4.1 Limitations of Simple Mean-Reversion (Part B)

Empirical Violations

1. Fat Tails (Excess Kurtosis)

$$\text{Kurtosis} = \frac{E[(r-\mu)^4]}{\sigma^4} \gg 3$$

Jarque-Bera test strongly rejects normality for commodity returns.

2. Autocorrelation in Residuals

$$\text{Corr}(\epsilon_t, \epsilon_{t-k}) \neq 0 \quad \text{for } k > 0$$

Box-Pierce Q-statistic significant: model underspecified.

1.4.1 Limitations of Simple Mean-Reversion (Part C)

Empirical Violations (continued)

3. Heteroscedasticity (Time-Varying Volatility)

$$\text{Var}(\epsilon_t|\mathcal{F}_{t-1}) = \sigma_t^2 \neq \sigma^2$$

ARCH-LM test: ϵ_t^2 shows significant autocorrelation.

4. Cannot Capture Spikes AR(1) with Gaussian errors cannot generate large, sudden price movements observed in data (especially electricity, natural gas).

Conclusion

Need more sophisticated models: GARCH, jump-diffusion, regime-switching.

1.4.1 Limitations of Simple Mean-Reversion (Part D): GARCH

GARCH(1,1) Model

Generalized Autoregressive Conditional Heteroscedasticity:

$$
\begin{aligned}
r_t &= \mu + \epsilon_t \\
\epsilon_t &= \sigma_t z_t, \quad z_t \sim N(0,1) \\
\sigma_t^2 &= \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2
\end{aligned}
$$

Interpretation:

- $\omega > 0$: Long-run variance level
- α: Reaction to recent shocks (ARCH effect)
- β: Persistence of volatility (GARCH effect)

Pedagogical clarification:

- Unlike AR(1)/OU: now volatility is time-varying
- Can generate clustering of high/low volatility periods
- Still Gaussian conditional on σ_t

1.4.1 Limitations of Simple Mean-Reversion (Part E)

GARCH Properties

Stationarity condition:

$$\alpha + \beta < 1$$

Unconditional variance:

$$E[\sigma_t^2] = \frac{\omega}{1 - \alpha - \beta}$$

Persistence: If $\alpha + \beta \approx 1$: Volatility shocks decay slowly (**integrated GARCH** or I-GARCH).

Advantages

- Captures volatility clustering
- Allows fat tails through time-varying volatility
- Widely used in risk management (VaR, CVaR)

1.4.1 Limitations of Simple Mean-Reversion (Part F): EGARCH

EGARCH Model (Nelson, 1991)

Exponential GARCH allows asymmetric effects:

$$\ln(\sigma_t^2) = \omega + \alpha \left(\left| \frac{\epsilon_{t-1}}{\sigma_{t-1}} \right| - \sqrt{\frac{2}{\pi}} \right) + \gamma \frac{\epsilon_{t-1}}{\sigma_{t-1}} + \beta \ln(\sigma_{t-1}^2)$$

Key feature: Leverage effect

- If $\gamma < 0$: Negative shocks increase volatility more than positive shocks.
- If $\gamma > 0$: Opposite (rare in equity markets, but can occur in commodities).
- No positivity constraints on parameters (log-volatility specification).

Pedagogical clarification:

- EGARCH addresses the limitation of GARCH for asymmetry
- Still conditional Gaussian but allows leverage effects
- Links to AR/OU as volatility-modulated mean-reversion models

1.4.1 Limitations of Simple Mean-Reversion: Summary

Comparison of Mean-Reverting Models for Commodities

Model	Key Features	Use / Limitations
AR(1)	Discrete-time, linear mean-reversion	Simple, short-term intuition; cannot capture volatility clustering or spikes
OU	Continuous-time analog of AR(1)	Analytical tractability; same limitations as AR(1)
GARCH(1,1)	Time-varying volatility; ARCH/GARCH effects	Captures volatility clustering and fat tails; Gaussian, no jumps
EGARCH	Log-volatility, asymmetric response	Models leverage effects, asymmetric spikes; widely used in commodities risk management

Key Takeaways

- AR(1)/OU: baseline models for teaching
- GARCH: realistic volatility structure
- EGARCH: asymmetric responses (electricity, natural gas)
- Model empirical violations: fat tails, heteroscedasticity, spikes

1.4.1 Limitations of Simple Mean-Reversion (Part G): Seasonal GARCH

Seasonal GARCH Model

Variance depends on past shocks and predictable seasonal effects:

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2 + \sum_{s=1}^{S} \delta_s D_{s,t}$$

Notation:

- σ_t^2: Conditional variance at time t
- ϵ_{t-1}: Previous period shock $(r_{t-1} - \mu)$
- $D_{s,t}$: Seasonal dummy (1 if period t in season s, 0 otherwise)
- δ_s: Effect of season s on volatility

Example: Natural gas volatility is higher in winter (heating demand) than in summer.

1.4.1 Seasonal GARCH (continued)

Estimation via Maximum Likelihood

Assuming normal shocks:

$$\max_{\theta} \sum_{t=1}^{T} \underbrace{-\frac{1}{2}\ln(2\pi)}_{\text{constant}} - \underbrace{\frac{1}{2}\ln\left(\sigma_t^2\right)}_{\text{volatility penalty}} - \underbrace{\frac{\epsilon_t^2}{2\sigma_t^2}}_{\text{fit to data}}$$

Goal: Find parameters $\theta = (\omega, \alpha, \beta, \delta_1, \ldots, \delta_S)$ that best explain observed returns.

Pedagogical Intuition

- **Why seasonality?** Commodities often have predictable volatility patterns (e.g., winter vs. summer for gas, summer peaks for electricity).
- **Effect of dummies:** $\delta_s D_{s,t}$ adjusts variance for each season, so the model "knows" when volatility is naturally high or low.
- **Benefit:** Improves forecasting, risk management, and pricing of options on commodities with strong seasonal behavior.

1.4.2 Modeling Price Spikes: Jump-Diffusion (Part A)

Merton Jump-Diffusion Model (1976)

Continuous price changes plus discrete jumps:

$$\frac{dS_t}{S_t} = (\mu - \lambda\kappa)\, dt + \sigma\, dW_t + (J_t - 1)\, dN_t$$

Notation (read left $\rightarrow$ right in the equation):

- S_t : Spot price at time t
- μ : Expected (unconditional) drift
- σ : Diffusion volatility (normal price fluctuations)
- W_t : Standard Brownian motion
- N_t : Poisson process with intensity λ (jump arrivals)
- J_t : Jump multiplier (typically lognormal)
- $\kappa = E[J_t - 1]$: Average proportional jump size

Drift adjustment: $-\lambda\kappa$ ensures $E[dS_t/S_t] = \mu$.

1.4.2 Modeling Price Spikes: Jump-Diffusion (Part B)

Interpretation

- **Diffusion** (σdW_t) $\rightarrow$ normal daily volatility
- **Jump timing** (dN_t) $\rightarrow$ when spikes occur
- **Jump magnitude** ($J_t - 1$) $\rightarrow$ size of spikes

Drift adjustment:

$$\mu - \lambda\kappa \rightarrow E\left[\frac{dS_t}{S_t}\right] = \mu$$

Ensures jumps do not artificially inflate expected growth.

Application

- Electricity: supply constraints $\rightarrow$ sudden upward spikes.
- Natural gas & storable commodities: spikes during peak demand.
- Useful for realistic modeling in risk management.

1.4.2 Mean-Reverting Jump-Diffusion (Part C1)

Mean-Reverting Jump-Diffusion Model

Combine mean-reversion with jumps in log-space:

$$d\ln(S_t) = \kappa(\theta - \ln(S_t))dt + \sigma dW_t + \xi_t dN_t$$

where $\xi_t \sim N(\mu_J, \sigma_J^2)$ and dN_t is a Poisson process with intensity λ.

Notations

- S_t: Commodity price at time t
- $\ln(S_t)$: Log-price (makes jumps multiplicative)
- κ: Mean-reversion speed — how quickly prices revert to equilibrium
- θ: Long-run equilibrium log-price
- σdW_t: Continuous normal (diffusion) fluctuations
- dN_t: Jump arrivals (Poisson, average frequency $= \lambda$)
- ξ_t: Jump size in log-space ($\sim N(\mu_J, \sigma_J^2)$)
- λ: Expected number of jumps per unit time
- Drift adjustment is implicit — ensures log-price tends to θ

1.4.2 Mean-Reverting Jump-Diffusion (Part C2): Intuition

Understanding the Dynamics

- **Diffusion term** (σdW_t): Normal day-to-day price fluctuations
- **Jump term** ($\xi_t dN_t$): Occasional spikes — sudden, large moves due to shocks
- **Mean-reversion** ($\kappa(\theta - \ln(S_t))$): After a spike, prices are pulled back toward equilibrium
- Combined effect: Prices mostly fluctuate around equilibrium but occasionally experience large, temporary deviations

Practical Example

- Electricity or natural gas prices: sudden spikes during peak demand.
- After spike, storage, market response, or supply adjustment brings prices back down.

1.4.2 Modeling Price Spikes: Jump-Diffusion (Part D)

Characteristic Function

For valuation and calibration, use the characteristic function:

$$\phi(u) = E\left[e^{iu\ln(S_T)}\right]$$

For the mean-reverting jump-diffusion model:

$$\phi(u) = \exp\left(iu\ln(S_0)e^{-\kappa t} + iu\theta(1-e^{-\kappa t}) - \frac{\sigma^2 u^2}{4\kappa}(1-e^{-2\kappa t}) + \lambda t\left(e^{iu\mu_J - \frac{1}{2}u^2\sigma_J^2} - 1\right)\right)$$

Notation

- i: imaginary unit in Fourier transforms to recover the probability distribution
- $\phi(u)$: Characteristic function of $\ln(S_T)$
- u: Fourier variable (complex argument)
- S_0: Initial commodity price
- κ: Mean-reversion speed of log-prices
- θ: Long-run equilibrium log-price
- σ: Diffusion volatility (continuous fluctuations)
- λ: Jump intensity (expected number of jumps per unit time)
- μ_J, σ_J^2: Mean and variance of jump sizes in log-space

1.4.2 Modeling Price Spikes with Jump-Diffusion: Intuition

Why Use Jump-Diffusion Models in Commodity Markets?

1. **Option Pricing**
   - Commodity options are sensitive to extreme price movements (spikes).
   - Jump-diffusion models generate heavier tails than Gaussian models.
   - Fourier methods allow fast and accurate pricing even with jumps.
2. **Market Calibration**
   - Standard diffusion models fail to fit observed implied volatility smiles.
   - Jumps help explain steep smiles and short-maturity skews.
   - Parameters can be inferred from option prices (risk-neutral jump intensity and size).
3. **Risk Management and Simulation**
   - Monte Carlo paths include rare but severe price spikes.
   - More realistic stress scenarios for VaR and CVaR.
   - Particularly important for electricity, gas, and power portfolios.

1.4.2 Modeling Price Spikes: Jump-Diffusion (Part E): Regime-Switching

Hamilton's Regime-Switching Model (1989)

Allow parameters to switch between discrete market states:

$$y_t = \mu_{s_t} + \phi y_{t-1} + \sigma_{s_t}\epsilon_t, \quad \epsilon_t \sim N(0,1)$$

where the latent state $s_t \in \{1, 2, \ldots, M\}$ follows a Markov chain with transition matrix P:

$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots \\ p_{21} & p_{22} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}, \qquad p_{ij} = \Pr(s_{t+1} = j \mid s_t = i)$$

Interpretation for Commodities

- s_t represents the **market regime** (e.g., normal vs stressed)
- Mean (μ_{s_t}) and volatility (σ_{s_t}) depend on the regime
- Regime switches are **persistent**, not iid (Markov structure)

1.4.2 Modeling Price Spikes: Jump-Diffusion (Part F)

Two-Regime Model for Commodities

Regime 1 (Normal market):

$$dx_t = \kappa_1(\theta_1 - x_t)dt + \sigma_1 dW_t$$

- Low volatility: σ_1 small
- Gradual mean-reversion: κ_1 moderate
- Captures typical day-to-day price dynamics

Regime 2 (Spike regime):

$$dx_t = \kappa_2(\theta_2 - x_t)dt + \sigma_2 dW_t + J_t dN_t$$

- High volatility: $\sigma_2 \gg \sigma_1$
- Fast mean-reversion: $\kappa_2 \gg \kappa_1$ (spikes decay quickly)
- Jumps occur: $\lambda > 0$ (supply shocks, outages)

1.4.2 Modeling Price Spikes: Jump-Diffusion (Part G)

Advantages of Regime-Switching Models

- Captures state-dependent volatility and dynamics
- Models abrupt transitions between calm and stressed markets
- More realistic than single-regime mean-reversion
- Identifies periods of structural change or stress

Estimation

Estimated via the **Hamilton Filter** (hidden-state filtering):

1. **Prediction**: $\Pr(s_t = i \mid \mathcal{F}_{t-1})$
2. **Update**: $\Pr(s_t = i \mid \mathcal{F}_t)$ using Bayes' rule
3. **Likelihood**: $L(\theta) = \prod_{t=1}^{T} f(y_t \mid \mathcal{F}_{t-1})$

Maximum Likelihood yields both parameter estimates and regime probabilities.

1.4.2 Modeling Price Spikes: Jump-Diffusion (Part H): Two-Factor Models

Schwartz-Smith Model (2000)

Log-price decomposed into short- and long-term components:

$$\begin{aligned} \ln(S_t) &= \chi_t + \xi_t \\ d\chi_t &= -\kappa\chi_t dt + \sigma_\chi dW_t^\chi \quad \text{(short-term deviations)} \\ d\xi_t &= \mu_\xi dt + \sigma_\xi dW_t^\xi \quad \text{(long-term equilibrium)} \\ dW_t^\chi dW_t^\xi &= \rho dt \end{aligned}$$

Notation and Interpretation

- χ_t: Transitory shocks (mean-reverting to zero)
- ξ_t: Persistent component (drifted random walk)
- κ: Speed at which short-term shocks dissipate
- ρ: Correlation between short- and long-term factors

1.4.2 Modeling Price Spikes: Jump-Diffusion (Part I)

Forward Curve in Schwartz-Smith Model

Forward price at time t for maturity T:

$$\ln F(t, T) = e^{-\kappa(T-t)}\chi_t + \xi_t + A(t, T)$$

where:

$$A(t, T) = \mu_\xi(T - t) + \text{variance/covariance adjustments}$$

Key Implications

- Short-term factor decays exponentially with maturity
- Long-term factor affects all maturities equally
- Spot spikes have limited impact on long-dated futures

Why This Model Is Used

- Jointly fits spot prices and entire forward curve
- Separates temporary shocks from structural trends
- Closed-form expressions (Gaussian factors)

1.4.2 Modeling Price Spikes: Jump-Diffusion (Part J): Gibson–Schwartz Model

Gibson–Schwartz Model (1990)

Joint model of spot price and convenience yield:

$$
\begin{aligned}
dS_t &= (\mu - \delta_t) S_t dt + \sigma_S S_t dW_t^S \\
d\delta_t &= \kappa(\alpha - \delta_t) dt + \sigma_\delta dW_t^\delta \\
dW_t^S dW_t^\delta &= \rho \, dt
\end{aligned}
$$

Notation and Economic Meaning

- S_t: Spot commodity price
- δ_t: **Convenience yield** (benefit of holding physical inventory)
- μ: Drift of spot price (before inventory effects)
- κ: Speed of mean-reversion of convenience yield
- α: Long-run average convenience yield
- σ_S, σ_δ: Spot price and convenience yield volatilities
- ρ: Correlation between price shocks and inventory shocks

Schwartz–Smith vs. Gibson–Schwartz: Comparison

- **Source of Mean-Reversion:**
  - Schwartz–Smith: Mean-reversion in short-term factor χ_t is statistical, not directly tied to inventory.
  - Gibson–Schwartz: Mean-reversion arises from **convenience yield** δ_t, reflecting physical inventory and storage decisions.
- **Economic Interpretation:**
  - δ_t directly measures **scarcity / storage effects**.
  - Negative correlation ρ is intuitive: Low inventory $\Rightarrow$ high δ_t $\Rightarrow$ upward pressure on S_t High inventory $\Rightarrow$ low δ_t $\Rightarrow$ less price pressure
- **Forward Curve Implications:**
  - Schwartz–Smith: Forward curve driven by latent short- and long-term factors.
  - Gibson–Schwartz: Forward curve shaped by stochastic convenience yield, giving a more physical/economic interpretation for futures pricing.
- **Pedagogical Takeaway:**
  - Use Gibson–Schwartz when explaining spikes and mean-reversion from a supply/storage perspective.
  - Use Schwartz–Smith when focusing on factor decomposition and analytical tractability for forward curves.

1.4.2 Modeling Price Spikes: Jump-Diffusion (Part K): Estimation

Maximum Likelihood for Jump-Diffusion

Log-returns $r_t = \ln(S_t/S_{t-1})$ under Merton jumps:

$$L(\theta) = \prod_{t=1}^{T} \sum_{n=0}^{\infty} \frac{e^{-\lambda \Delta t}(\lambda \Delta t)^n}{n!}$$

$$\times \frac{1}{\sqrt{2\pi(\sigma^2 \Delta t + n\sigma_J^2)}} \exp\left(-\frac{(r_t - \mu \Delta t - n\mu_J)^2}{2(\sigma^2 \Delta t + n\sigma_J^2)}\right)$$

Reading the Likelihood

- Returns arise from diffusion plus a random number of jumps
- n = number of jumps in the period
- Poisson weights reflect jump probabilities

Practical note:

- Infinite sum reflects uncertainty over the number of jumps
- In practice, truncate at $n_{\max}$ (e.g., 10)
- Likelihood approximated due to fast decay of Poisson probabilities

Kalman Filter: Measurement and State Equations

Purpose

Estimate latent factors in Schwartz-Smith and similar state-space models.

Measurement equation

$$y_t = Z\alpha_t + d + \epsilon_t, \quad \epsilon_t \sim N(0, H)$$

- y_t: Observed data (spot prices or log-prices)
- α_t: Latent factors (short-term χ_t, long-term ξ_t)
- Z: Factor loadings linking α_t to observations
- d: Deterministic shift
- ϵ_t: Measurement noise

State equation

$$\alpha_t = T\alpha_{t-1} + c + R\eta_t, \quad \eta_t \sim N(0, Q)$$

- T: Factor transition matrix
- c: Drift term in factor evolution
- R: Factor shock loadings
- η_t: State noise

Kalman Filter: Likelihood and Practical Use

Likelihood via Prediction Error Decomposition

$$L(\theta) = \prod_{t=1}^{T} f(y_t | \mathcal{F}_{t-1})$$

- Prediction errors are Gaussian
- Kalman filter recursively computes means and variances of α_t

Practical Use

- Estimate latent short-term and long-term factors from observed prices
- Update beliefs on market dynamics as new data arrives
- Feed latent factors into forward curve modeling or spike simulations
- Provides analytical tractability for option pricing and risk analysis

1.4.2 Modeling Price Spikes: Jump-Diffusion (Part M): Model Selection

Information Criteria for Model Selection

Balance goodness-of-fit vs. model complexity:
Akaike Information Criterion (AIC):

$$AIC = -2 \ln L + 2k$$

Bayesian Information Criterion (BIC):

$$BIC = -2 \ln L + k \ln n$$

Hannan-Quinn (HQIC):

$$HQIC = -2 \ln L + 2k \ln(\ln n)$$

Notation / Intuition:

- L: Maximum likelihood of the model
- k: Number of estimated parameters
- n: Sample size
- Lower value $\Rightarrow$ better trade-off between fit and parsimony
- BIC penalizes complexity more heavily than AIC

1.4.2 Modeling Price Spikes: Jump-Diffusion (Part N): Forecast Evaluation

Diebold-Mariano (DM) Test

Compare predictive accuracy of two models:

$$DM = \frac{\bar{d}}{\hat{\sigma}_{\bar{d}}/\sqrt{n}} \sim N(0,1)$$

Notation / Intuition:

- $d_t = L(e_{1t}) - L(e_{2t})$: Loss differential at time t
- L: Loss function (e.g., MSE, MAE)
- $|DM| > 1.96 \Rightarrow$ Significant difference at 5% level
- Sign of DM $\Rightarrow$ which model performs better

Common Loss Functions

- MSE: $L(e_t) = e_t^2$ (emphasizes large errors)
- MAE: $L(e_t) = |e_t|$ (robust to outliers)
- QLIKE: $L(e_t) = \frac{e_t^2}{\hat{\sigma}_t^2} + \ln\left(\hat{\sigma}_t^2\right)$ (for volatility forecasting)

1.4.2 Modeling Price Spikes: Jump-Diffusion (Part O): VaR Backtesting

Value-at-Risk (VaR)

VaR at confidence level α is the loss threshold exceeded with probability $1 - \alpha$:

$$\Pr(\text{Loss} > \text{VaR}_\alpha) = 1 - \alpha$$

Example: 99% daily VaR = \$1M $\Rightarrow$ 1% chance of losing more than \$1M in one day.

Kupiec's Unconditional Coverage Test

Check if observed violation frequency matches expected:

$$LR_{\text{uc}} = -2\ln\left[(1-p)^{T_0} p^{T_1}\right] + 2\ln\left[\left(1 - \frac{T_1}{T}\right)^{T_0}\left(\frac{T_1}{T}\right)^{T_1}\right] \sim \chi^2(1)$$

Notation / Intuition:

- T_0: Number of days without VaR violation
- T_1: Number of days with VaR violation
- $T = T_0 + T_1$: Total sample size
- $p = 1 - \alpha$: Expected violation rate
- Test statistic $\sim \chi^2(1)$ under null hypothesis

Synthesis of Models for Commodity Prices

Model	Mechanism	Economic Interpretation	Key Parameters / Use Case
Jump-Diffusion (Merton)	Continuous diffusion + occasional jumps	Captures sudden spikes from supply shocks, weather, or other rare events	λ (jump frequency), ξ_t (jump size), σ (diffusion) Use case: short-term spike risk, option pricing, Monte Carlo
Regime-Switching (Hamilton)	Parameters switch between discrete states	Models "normal" vs "stress/spike" regimes, captures volatility clustering	p_{ij} (transition probabilities), σ_i, κ_i Use case: detect stress periods, discrete jumps
Two-Factor Models (Schwartz–Smith, Gibson–Schwartz)	Latent short-term and long-term factors; optionally stochastic convenience yield	Short-term factor captures transitory deviations; long-term factor captures equilibrium trend; can link mean-reversion to inventory	χ_t/ξ_t or S_t/δ_t, correlation ρ, mean-reversion κ Use case: forward curve modeling, separates short-run volatility vs long-term trend.

Lecture Conclusion

Pedagogical Takeaways

- **Jump-Diffusion** captures sudden, extreme price moves—critical for short-term risk and derivative pricing.
- **Regime-Switching** models discrete market states—good for stress testing and capturing volatility clustering.
- **Two-Factor models** explain both short-term fluctuations and long-term trends—connects to inventory, storage, and convenience yields.
- **Practical insight**: In real commodity markets, combining approaches (e.g., two-factor + jumps) often provides the most realistic modeling.
- Always align model choice with empirical observations: spikes, mean-reversion, forward curve shape, and storage effects.

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Thank You

Questions?

Part II: Forward Curve & Multi-Commodity Modeling
Coming next...